A monitoring desk for human review

Current software scope and honest evaluation boundaries

The prototype lets an operator inspect person and vehicle events, mark alerts reviewed, filter the queue and export a session. A local Python service accepts validated events from an optional YOLO adapter.

Implemented software

Offline sample session, local HTTP service, event review and JSON export.

Optional inference path

Local video/webcam adapter. Requires packages and trusted weights; actual inference not yet run.

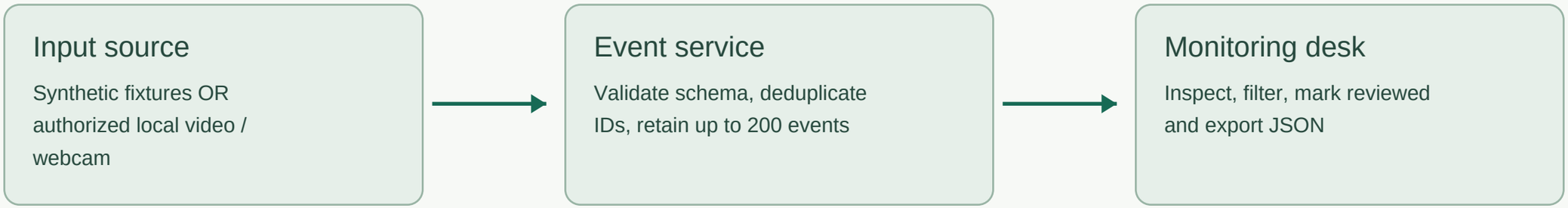
Aircraft control, GPS missions, MQTT and browser video streaming are proposed future integration. No aircraft was connected. Synthetic sample imagery is illustrative and must never be used as detection evidence.

Validation: 13 Python tests and the offline UI workflow harness pass.

Live networking and rendered browser QA were blocked by the build sandbox.

Software architecture

Implemented local event path; aircraft system remains independent



Real video passes through the optional YOLO adapter before event ingestion.

Proposed aircraft layer

Pixhawk: stabilization and configured failsafes. Raspberry Pi: companion processing. Hardware unverified.

Separation of responsibilities

No flight-control endpoint exists in the prototype. Detections require human review; no automatic pursuit.

Actual interface: loopback HTTP JSON. MQTT and remote video are not implemented.

Detection and event flow

Validation, bounded storage and review states

1. Read an authorized frame

Optional YOLO adapter; timestamp is frame-read time.

2. Filter supported classes

Confidence threshold and per-class cooldown.

3. Validate the JSON event

Types, UTC time, confidence, normalized box and source ID.

4. Deduplicate and retain

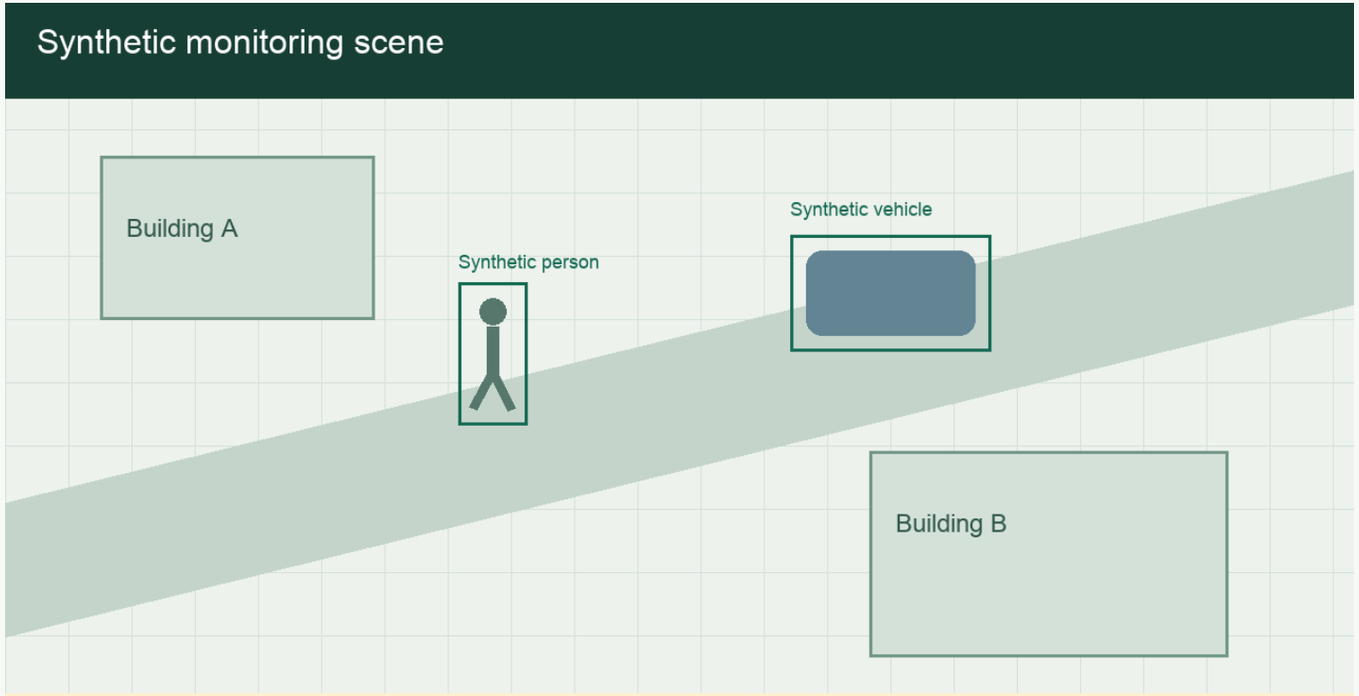
Latest 200 events in memory; reset on server restart.

5. Review and export

Operator inspects the event and marks it reviewed.

Dashboard wireframe

Illustrative inspection scene with a separate review queue



Review queue

Person: synthetic sample. Inspect its normalized box. Mark reviewed. Filter pending events. Export JSON.

Flight telemetry

Not connected. Battery and GPS unavailable.

The browser scene is an illustration. Original camera frames appear only in the optional OpenCV preview window. The UI must not imply that the illustration is a live video stream.